

Full length article

An investigation on microstructure and mechanical properties of H62 brass thin-sheet by fiber laser welding: Experiments and multi-scale simulations

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ARTICLE INFO

Keywords:

H62 brass
Laser welding
Mechanical properties
Multi-scale simulations

ABSTRACT

H62 brass is widely used as an engineering material in electrical components, automotive parts, aerospace and other fields. Many brass products and structures requiring material forming by means of welding. The reliability of the welding process directly affects the quality of the brass welded joints. In this study, a high-power fiber laser system was used for welding tests on thin sheets of H62 brass. Based on the theory of a single control variable with multiple welding parameters, the laser power, the amount of defocusing and welding speed were investigated in relation to the relationship of joint formation, microstructure and mechanical properties. In addition, two kinds of macroscopic segregations were found in some welded joints due to insufficient heat input and subcooling of the melt pool to the submixing gap. In order to further analyze the forming mechanism during the welding process, the physical model was established by finite element and molecular dynamics research methods. Using finite elements, the influence rule of different process parameters on the temperature field was analyzed; molecular dynamics was used to analyze the formation process of the weld from the atomic scale. Ultimately, this study has resulted in welded joints of H62 brass with excellent mechanical properties and no obvious defects through suitable combinations of different process parameters. Most importantly, a multi-scale simulation model has been established to analyze the rules of temperature influence in the weld formation process.

1. Introduction

Brass is now considered a favorable engineering material to be applied to industry due to its advantages, including high strength, high corrosion resistance and high thermal conductivity [1]. In recent years, based on industrial development and technological progress, brass welded joints require superior performance. In particular, when brass is used in industrial production as the metal side of sandwich panels, the bending properties will become an important indicator for welded joints [2–4]. Laser welding has the advantages of small functional area, rapid heating, rapid cooling, etc. The use of laser welding helps to achieve high quality welded joints, it has been acknowledged that the in-depth research on laser welding of brass holds great theoretical and engineering values [5–7].

There have been some researches on laser welding of brass. Some scholars found that the laser welding process for H62 brass resulted in a

depressed weld after welding due to partial evaporation of zinc and spattering of copper [1,8,9]. Wang et al. [10] analyzed the structural characteristics of H62 brass welds. The fusion zone near the base material side is mainly composed of columnar crystals, while near the centre of the weld is mainly composed of fine equiaxed crystals, and the centre of the weld is completely composed of fine grains. Wang et al. [11] studied the influence of process parameters on weld formation, internal defects and alloy composition changes during laser welding of H62 brass, and emphasized that heat input to the welding pool must be controlled during the welding process. In addition, some studies have focused on laser welding of H62 brass and dissimilar alloys [12–15]. Dong et al. [12] studied the laser lap welding of copper and brass, they found that brass has a low threshold of deep penetration due to its high zinc content and low evaporation point, which makes it easy to achieve fusion penetration in laser welding. Ding et al. [13] researched the effect of process parameters on weld shape and temperature distribution in

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<https://doi.org/10.1016/j.optlastec.2023.110376>

Received 27 June 2023; Received in revised form 14 October 2023; Accepted 20 November 2023

Available online 3 December 2023

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heterogeneous laser welding of stainless steel 308 and brass alloys, and found that increasing weld pool size (width and depth) by increasing peak power and decreasing welding speed is more effective than other parameters. Li et al. [14] investigated the effect of the pulse duration and peak power of Nd:YAG pulsed laser beam on the microstructure and mechanical properties of H62 brass and 316L stainless steel welded joints under constant pulse energy. Zhou et al. [15] in order to establish the thermal cycle at the interface and thus better understand and elucidate the microstructural evolution of the reaction products. Numerical simulations of the laser brazing process of Al/brass dissimilar joints were carried out to obtain temperatures at the interface that are difficult to measure directly by thermocouples. Aghaee et al. [16] successfully predicted the weld geometry, temperature distribution and strain distribution of a laser welded joint between copper and 304 stainless steels by numerical simulation. Feng et al. [17] investigated the bonding process of Cu and Fe in explosive welding by molecular dynamics simulation, and analyzed the morphologies and atomic structures of the three bonding interfaces in each stage. Badawy et al. [18] developed a multi-scale model for laser welding by coupling the finite element method and molecular dynamics. The temperature range and the crystal structure within and near the welded pool could be predicted by the proposed model.

Overall, the previous research on laser welding of brass and its alloys has explored various aspects, including weld formation, structural characteristics, defects, changes of alloy composition, and even the welding of dissimilar alloys. These studies have identified the need for controlling the laser parameters during welding and optimizing the welding process to achieve better weld quality. However, more research is needed to fully understand the complex mechanisms involved in laser welding of brass and its applications, especially in industry.

The research proposed in this paper aims to fill the gap in the understanding of the welding forming rule in laser welding of brass. By using the laser welding process and the control variables method to prepare H62 brass sheet specimens, this study will analyze the influence law of laser power, the amount of defocusing and welding speed on joint forming, microstructure morphology, and mechanical properties. Additionally, a physical model of laser welding of brass will be established using finite element method to further analyze the forming law of laser welding of H62 brass and obtain the influence law of different process parameters on the temperature field. In order to obtain the atomic scale evolution of weld formation during the welding process, molecular dynamics simulations are conducted to reproduce the Cu-Zn weld joining process from the atomic level.

This study will contribute to the field of laser welding of brass by providing new insights into the welding forming rule and identifying reasonable process parameters. The use of simulation methods will also allow for more accurate predictions of the effects of different process parameters on the welding process, which can ultimately improve the quality and efficiency of brass laser welding in industry.

2. Experimental procedure

The base material used in this experiment was H62 brass, which is a two-phase brass composed of α -phase and β -phase at room temperature. The dimensions of brass sheet were $100 \times 50 \times 2 \text{ mm}^3$. Both sheets were placed horizontally and a 100 mm length of edge was butt-laser welded without leaving a gap between the welded parts. The chemical composition and mechanical properties of H62 brass are shown in Table 1 and Table 2.

Fig. 1 shows the schematic diagram of laser welding of brass. This

Table 1
Chemical compositions of H62 copper alloy.

Composition	Cu	Fe	P	Pb	Zn	Impurities
% Weight	61.2	0.08	0.004	0.0014	Bal.	0.23

Table 2
Mechanical properties of H62 copper alloy.

σ_b (Mpa)	δ (%)	Hardness(HV)	Bending strength(MPa)
525	28	125	631.4

experiment used a MFSC – 4000 w continuous fiber laser at a wavelength of 1080 nm with a rated power of 4000 w and an output fiber radius of 200 μm .

The surface of the specimen was polished and cleaned with alcohol before welding. During welding, the sample was fixed on the workbench with fixtures, argon was used as the protective gas, and the flow rate of the top blowing argon (coaxial with the laser beam, blowing towards the front of the groove) and the flow rate of the trailing cover argon (protecting the hot weld) were both 20 L/min. The back of the groove was covered on the slot, the flow rate of argon in the slot is 10 L/min. And the beam axis Angle was 15° . In this research, three primary variables in the laser welding process will be investigated: laser power, the amount of defocusing, and welding speed [19]. Based on the data from several pre-experiments, the combination of welding process parameters was determined, as shown in Table 3.

The welded specimens were cut using wire cutting equipment to obtain a cross section perpendicular to the direction of the weld. Standard metallographic sample was prepared and etched with $\text{FeCl}_3 + \text{HCl}$ solution ($\text{FeCl}_3:\text{HCl}:\text{H}_2\text{O} = 5 \text{ g}:10 \text{ ml}:100 \text{ ml}$) for 10 s. The microstructures were then observed using optical microscope (OM) and scanning electron microscope (SEM). The images taken by SEM include secondary electron imaging (SE) mode and backscattered electron imaging (BSE) mode. The chemical composition distribution was analyzed using energy dispersive spectrometer (EDS) fitted to the SEM.

The FEM software (*COMSOL Multiphysic*) was used to calculate the transient thermal field in the welding process, and then to study the effect of different welding process parameters on the temperature field. The atomic scale simulation of the weld formation process was simulated by *LAMMPS package* and the data generated during the molecular dynamics simulation were visualised and analysed by *Ovito*.

A micro-Vickers hardness tester (THV-1MT) was used to measure the hardness of the specimen. The load was set at 500 g, and the pressure holding time was 10 s. Points were taken at 0.8 mm below the upper surface of the weld at an interval of 0.1 mm. QJ211S universal mechanical testing machine was used for the bending test of welded joint. The thickness of the bending sample was 2 mm, the width was 5 mm, the span was 24 mm, and the bending speed was 1 mm/min.

3. Results and discussion

3.1. Microstructure analysis of laser welded joint

Fig. 2 shows the optical images and SEM images of the section of the welded joint. It can be seen that the weld penetrates the entire section, typical of a deep laser fusion weld. There are varying degrees of “ depressions ” on the front and back of the weld, which can be attributed to the evaporation of Zn elements due to laser action [9]. The entire section is divided into the weld zone (WZ), the fusion zone (FZ), the heat affected zone (HAZ) and the base material (BM). The central part of the WZ consists of equiaxed crystals and a small number of columnar crystals. This was due to the uniform temperature distribution, small temperature gradient and fast cooling rate in the middle of the WZ. Therefore, it was easy to generate fine equiaxed crystals during the welding process. The morphology of the FZ is mainly columnar, accompanied by a few equiaxed crystals. During welding the FZ was subjected to the cooling effect of the base material, resulting in a large temperature gradient, which contributed to the growth of columnar crystals. In addition, the β -phase is found in the FZ, which may be the β -phase introduced by the stirring of convection in the molten pool [20]; the β -phase is also found in the weld zone, which may be caused by

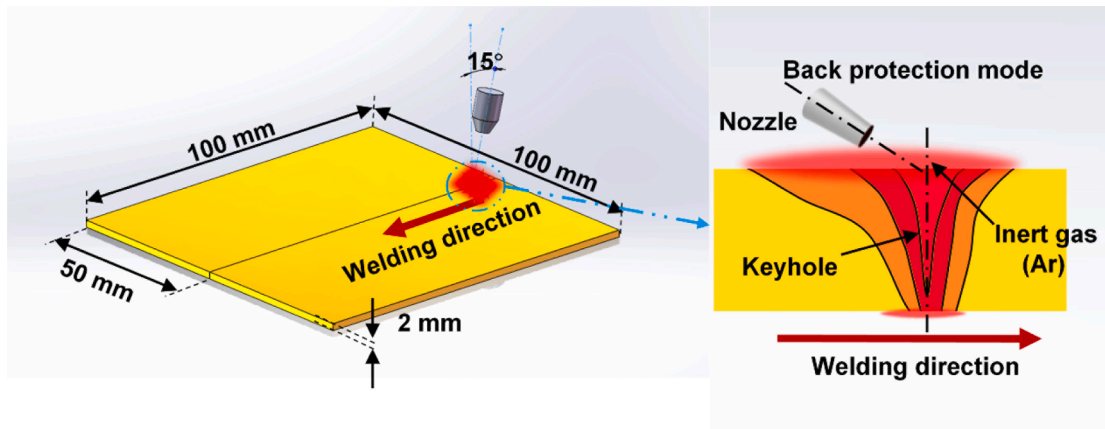


Fig. 1. Schematic diagram of laser welding of H62 brass.

Table 3

The used laser parameters in the experiments.

Exp (No.)	Laser power (W)	The amount of defocusing(mm)	Welding speed (mm/s)
1	1700	0.5	15
2	1800	0.5	15
3	1900	0.5	15
4	2000	0.5	15
5	1800	0	15
6	1800	1	15
7	1800	1.5	15
8	1800	0.5	11
9	1800	0.5	13
10	1800	0.5	17

segregation, the reasons for which are discussed in detail in the next section.

The microstructure of the HAZ is shown in Fig. 3. The constituent phases of the HAZ are mainly α -phase and β -phase. α phase is a solid solution with Cu as matrix and Zn dissolved in it. The crystal structure of α phase is face-centered cubic (FCC), which has low strength and good plasticity [21]. The β phase is a CuZn-based solid solution, and its strength and hardness are improved compared with the α phase, because its crystal structure is body-centered cubic (BCC). The plasticity of β phase is poor, but it can be improved at high temperature. The β -phase in HAZ was finely dispersed and formed fine crystal structure. This indicates that it was heated more evenly during the welding process.

3.2. Macrosegregation in laser welded joints

3.2.1. Macrosegregation caused by copper droplet and zinc vapor

In several parameter combinations with relatively low heat input, such as No. 3 and No. 6, the area shown in Fig. 4-a is found near the top of the weld. The images of SEM reveal the presence of weld microholes, cavities, etc. in the area, the most conspicuous of which is a “crescent” structure. The α phase, the β phase and the disordered tissue are distributed close to the edge of the “crescent”. There are two distinct zones within the “crescent” and the junction between them could easily become the site of cracking. The distribution of elements in this area was analyzed by EDS, as shown in Fig. 4-b, which shows that the central part of the “crescent” is copper-rich and the two sides are zinc-rich.

During laser welding, the laser evaporates a keyhole that acts as a heat source in the thickness direction. The keyhole is kept open by the recoil pressure of the zinc vapour. Due to its high energy density, the fiber laser has an excellent potential to form many sputtering particles during the welding process [22–24]. The copper droplet breaks free from the molten pool at the keyhole and forms a splash, which requires

overcoming the escape threshold. The escape threshold consists of the combination of local values: surface tension, surface geometry, viscosity and gravity, of which surface tension is the most important consideration [25,26]. If the driving force provided by the laser heat source and the zinc vapour to the copper droplet was not sufficient to overcome its escape threshold, the copper droplet would be closed in the keyhole. In addition, there may be zinc vapour in the keyhole that had not escaped in time when the hole was closed. Upon cooling, molten copper droplets that have not yet escaped from the keyhole form copper-rich regions; The trapped zinc vapor forms zinc-rich regions and cavities. This is the reason for the formation of the “crescent” near the top of the weld.

3.2.2. Macrosegregation caused by constitutional supercooling

Laminar structures are found on the backs of the welds for almost all combinations of parameters, as shown in Fig. 5. Some equidistant points were introduced into the structure, and the distribution of elements at these points was obtained by EDS. The laminar structure is strongly correlated with the distribution of the elements. This distribution of elements produced the constitutional supercooling, which would influence the way of growth during solidification and crystallisation, leading to the formation of the laminar structure. During the welding process, the cooling rate of laser welding was quite fast, enough to undercool the bulk molten pool into the metastable miscibility gap of the Cu-Zn phase diagram, forming a Zn-rich liquid phase and a Cu-rich liquid phase, and the former as spheres and the latter as matrix [27]. As the zinc-rich liquid is lighter than the copper-rich liquid, the zinc-rich spheres float upwards to the top of the copper-rich liquid. The viscosity of the liquid metal would increase significantly with the increase of the degree of supercooling, and the supercooled bulk welding pool may also be viscous, which can resist mixing of the them [27,28]. At the same time, the higher cooling rate could help the layered liquid to solidify before dispersion. Therefore, these factors promoted the formation of layered structures in the molten pool.

3.3. Finite element simulation

3.3.1. Numerical model and its verification

In this study, the software of **COMSOL Multiphysic** was used for finite element simulation, the selected heat source was a combination of Gaussian surface heat source and cylinder heat source [29,30]. The entire transient and non-linear heat transfer was analyzed based on the heat conduction equation. The Gaussian surface heat source was used for the surface part of the weld:

$$q_1(r) = \frac{3Q\eta_1}{\pi r_1^2} \exp\left(-\frac{3r^2}{r_1^2}\right) \quad (3-1)$$

Cylindric heat source was used in other parts of welding. The

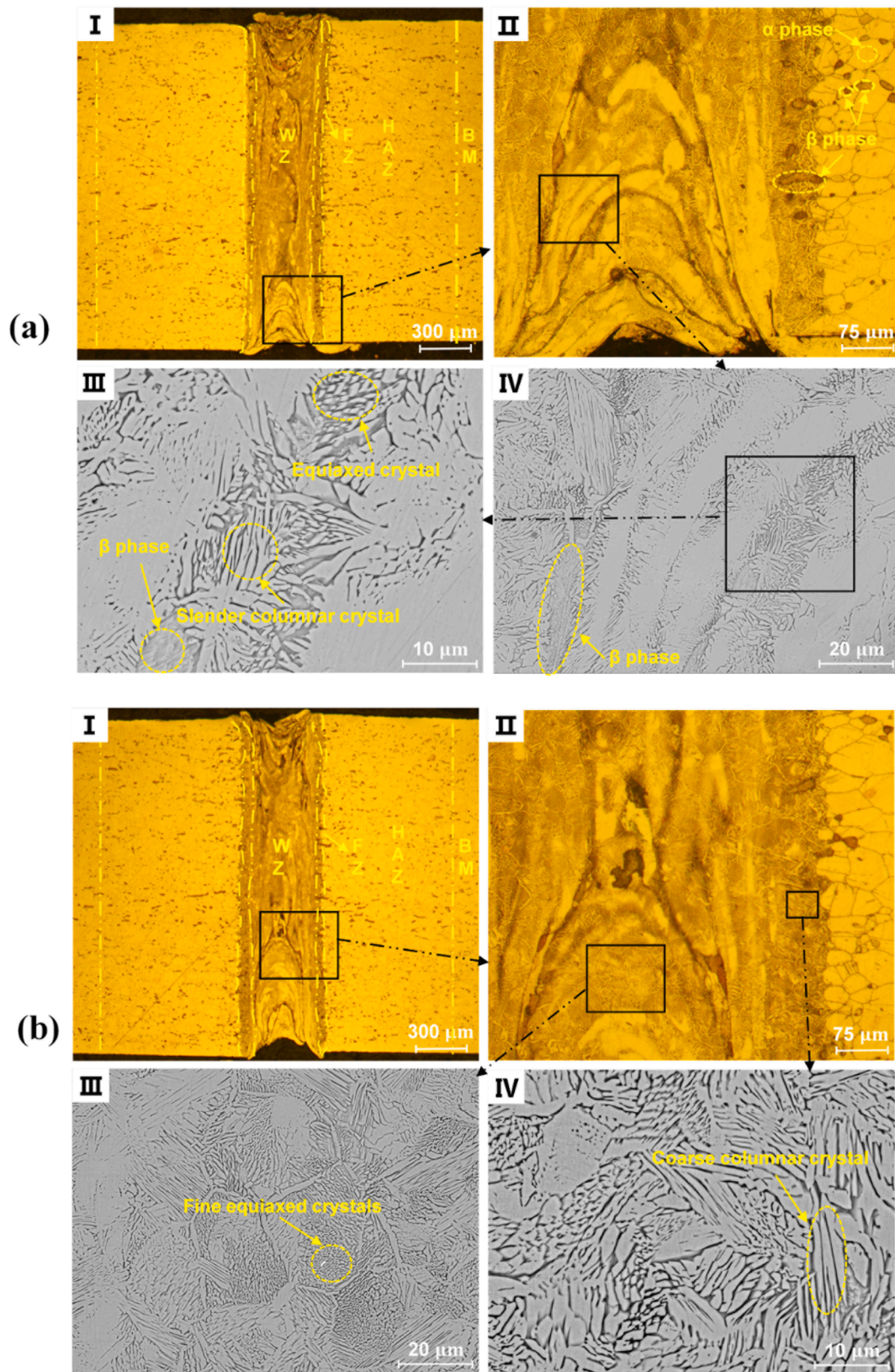


Fig. 2. Microstructure of different welding parameters: (a) laser power = 1800 W, the amount of defocusing = 1 mm, welding speed = 15 mm/s; (b) laser power = 1900 W, the amount of defocusing = 0.5 mm, welding speed = 15 mm/s.

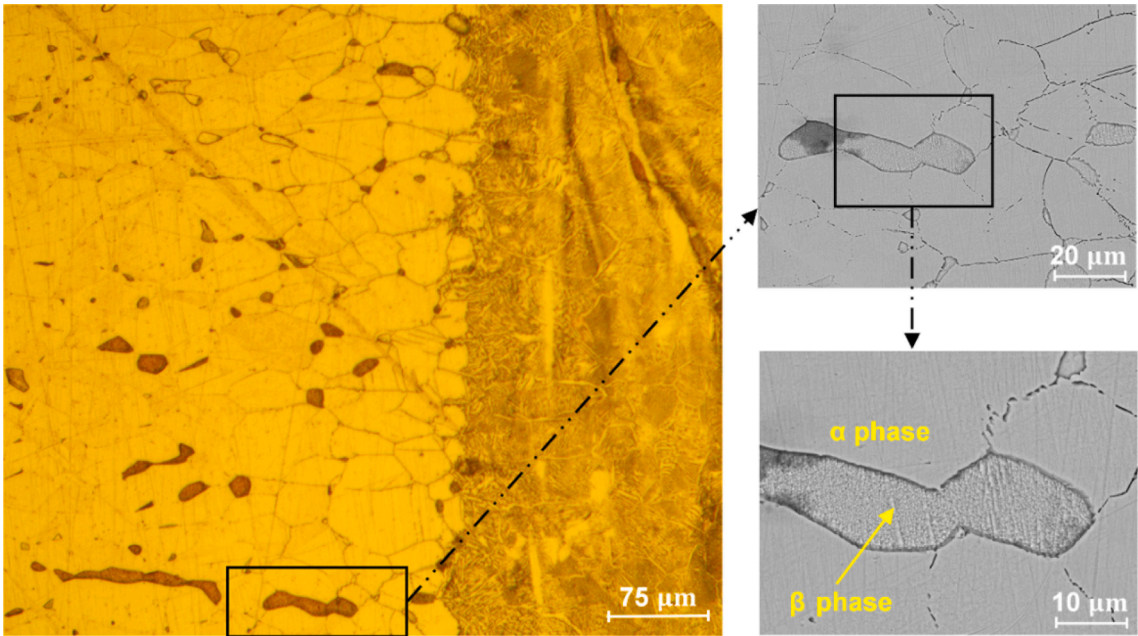


Fig. 3. Microstructure of heat affected zone.

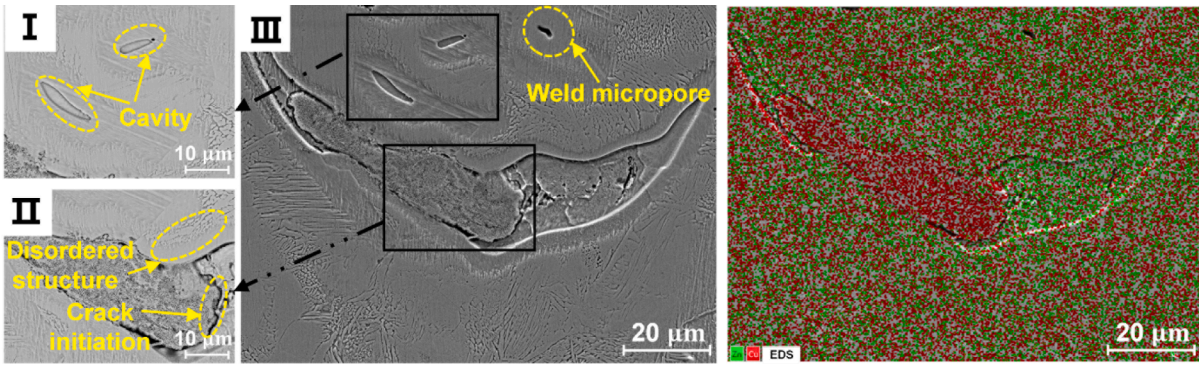


Fig. 4. The area near the top of the weld: (a) SE and BSE; (b) EDS.

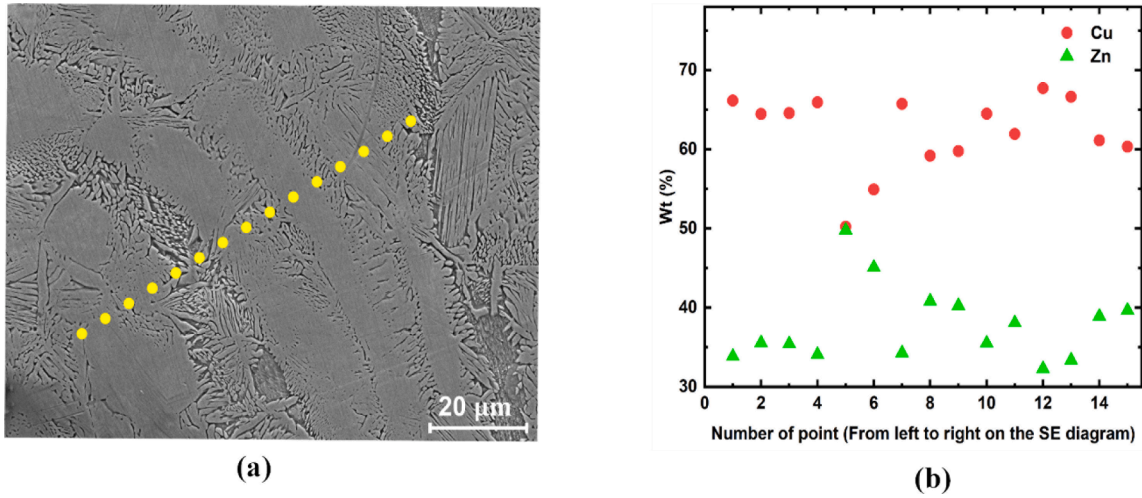


Fig. 5. (a) The layered structure, (b) The distribution of elements.

cylindric heat source model also presents Gaussian energy distribution in the radial direction, but linearly decreases in the thickness direction:

$$q_2(r, z) = \frac{6Q\eta_2}{\pi r_2 H (mH + 2r_2)} \exp\left(-\frac{3r^2}{r_2^2}\right) \left(\frac{mh + r_2}{r_2}\right) \quad (3-2)$$

Where Q is the power (W) of energy input during welding; η_1 and η_2 are the proportion of Gaussian surface heat source and cylinder heat source to the whole welding heat source, respectively; r is the distance (mm) from the center of the molten pool; r_1 and r_2 are the concentration coefficients of Gaussian and cylindrical heat sources, respectively; H is the height of cylindric heat source (m); h is the position of cylindric heat source in the direction of thickness; m is the energy regulating parameter of the upper and lower surfaces of the cylinder heat source. The heat source model was defined using the trial-and-error method in order to get a better match between the simulation results and the experimental results.

The laser welding process was simulated by using laminar flow and fluid heat transfer modules in non-isothermal flow field. Based on the actual research, in order to save computing resources and obtain more accurate simulation results, the 1/4 model of the actual weld (2D symmetric model of the upper part of the weld) was established. The upper part of this is the gas phase region and the lower part is the liquid and solid phase region, as shown in Fig. 6-a. In the geometric model, the upper part of the weld area is finely meshed and the lower part is finer to ensure a more accurate simulation, while the meshes away from the weld area are coarser to save calculation time. All meshes are free

triangular meshes. Fig. 6(b-d) shows the comparison between experimental and simulated depression depth at different welding process parameters. The depression depth and variation trend of the two cases are basically consistent, the simulation results and experimental results agree well with each other.

3.3.2. Influence of different process parameters on temperature field

Obtaining optimal conditions for the welding process in laser welding has always been a challenge. Copper and copper alloys exhibit low process stability during welding due to their unstable melt pool behavior, so extensive testing is required to investigate the effect of different process parameters [31–33]. In this research, finite element simulations were used to study the effects of laser power, amount of defocusing and welding speed on the weld temperature field. In the following part, if not specified, the default laser power is 1800 w, the amount of defocusing is 0.5 mm and the welding speed is 15 mm/s. The two-dimensional mirrored temperature fields of the welded joint at 40 ms under different parameters are extracted and shown in Fig. 7.

As shown in Fig. 7-a, the maximum temperature increases as the power increases, and the high temperature zone slowly expands from the opening of the small hole to the inside of the hole until it penetrates through the entire hole. This would evaporate more zinc and provide more driving force to the droplets inside the hole. This is also why the segregation caused by the droplet did not appear in the process parameter combination with higher heat input.

As shown in Fig. 7-b, with the increase of defocus, the laser spot on

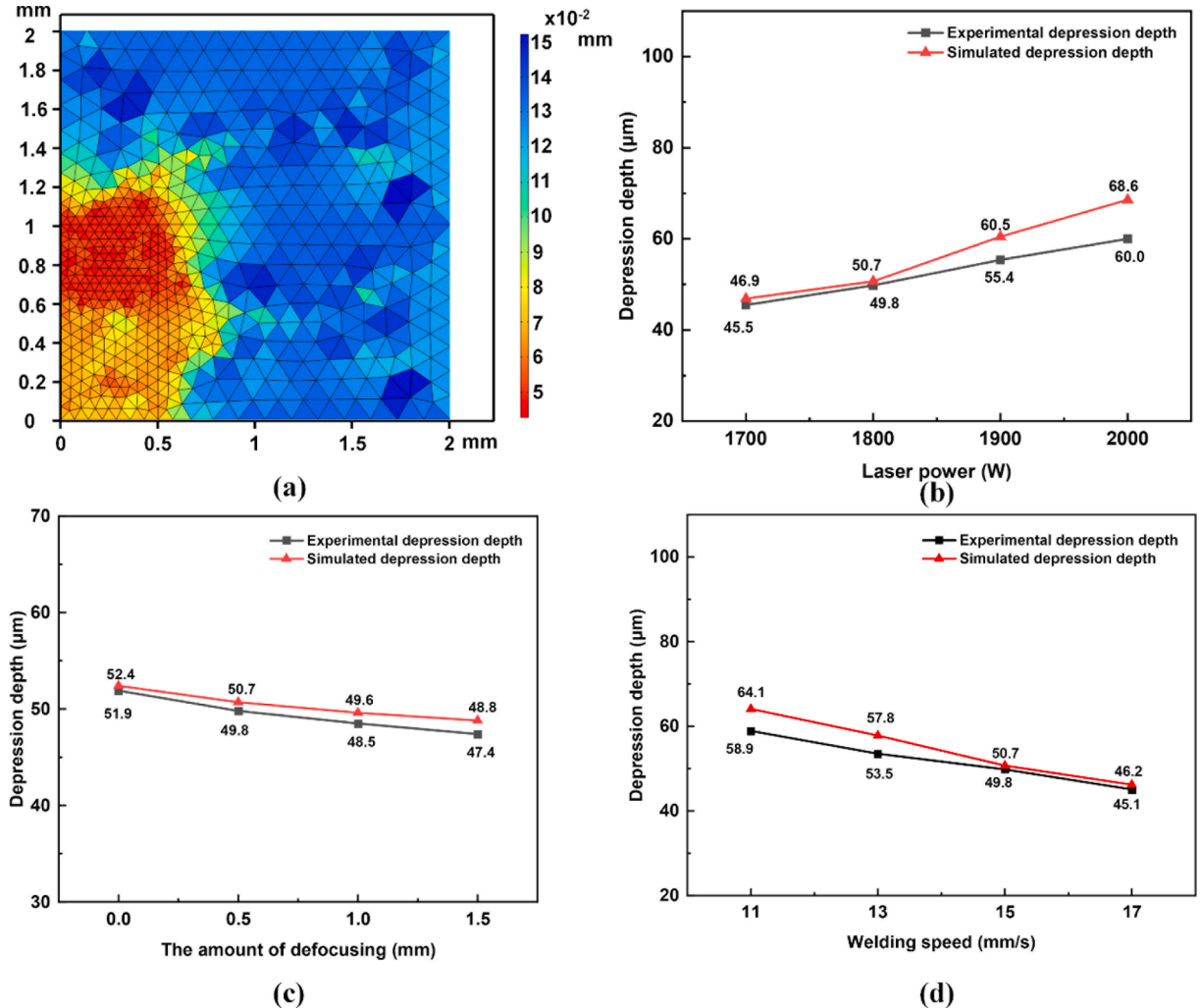


Fig. 6. (a) Meshing of the model and mesh size; (b-d) The Experimental depression depth and simulated depression depth of different welding process parameters.

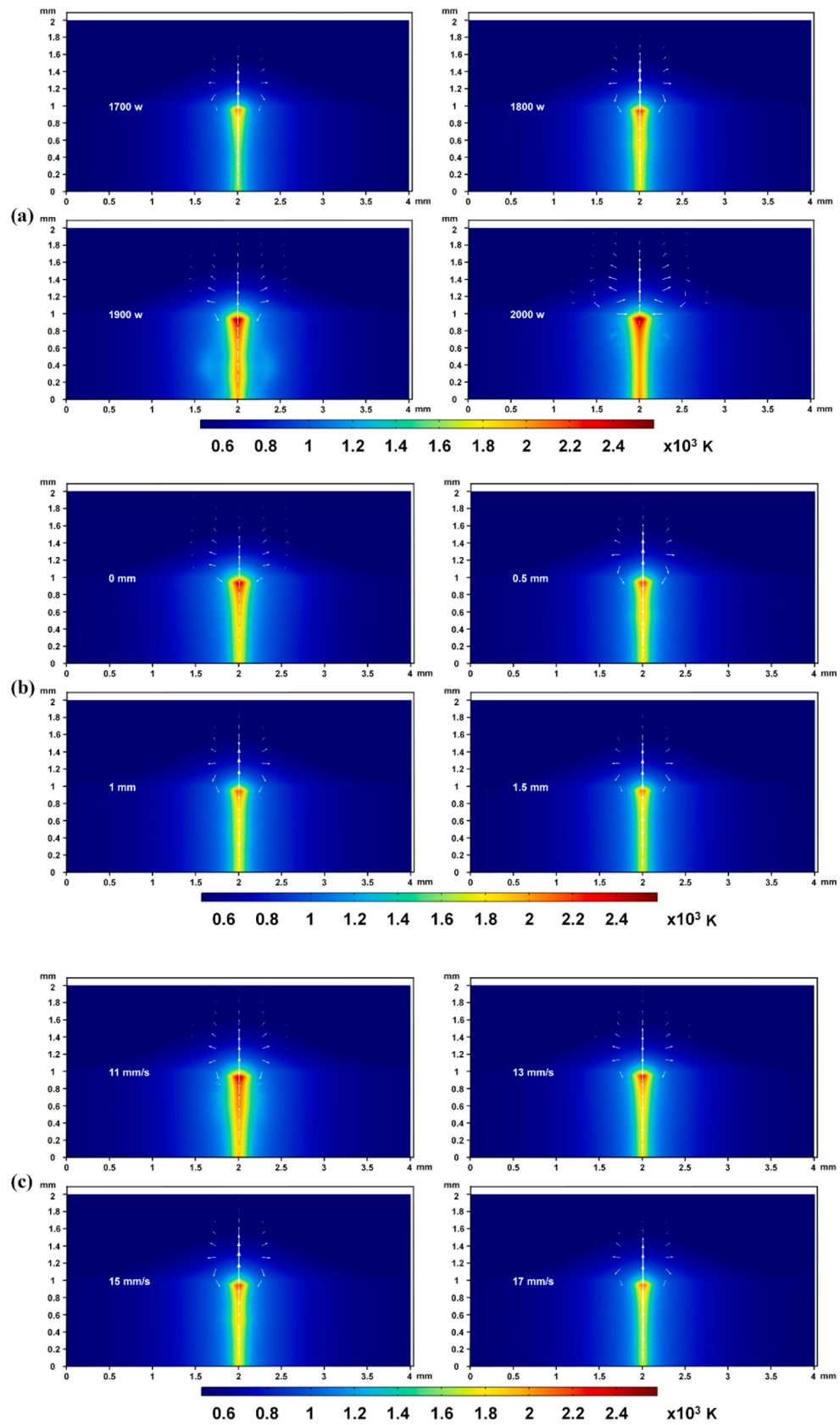


Fig. 7. Temperature field under different parameters ($t = 40$ ms)-, (a) Laser power (w):1700、1800、1900、2000; (b) The amount of defocusing (mm):0、0.5、1、1.5; (c) Welding speed (mm/s):11、13、15、17.

the surface of the weld specimen becomes larger, and the laser energy becomes more dispersed. This reduces the heat input at the weld center and narrows the area of high temperature. In addition, with the increase of defocusing amount, the degree of its influence on the temperature field gradually decreases.

As shown in Fig. 7-c, the welding speed can affect the energy absorption time of the welding area, thus affecting the temperature field. With welding speed increasing, the heating time becomes relatively shorter and the high temperature region and maximum temperature decrease. When the welding speed is at a relatively low value, and the change of welding speed has more obvious influence on the temperature field.

In summary, the process parameters are closely related to the heat input [34]. In the case of other process parameters remain constant, changing the single process parameter. The increase of laser power will increase the laser power density on the welded sample, thus improve the heat input. The amount of defocusing is directly related to the power density on the workpiece. With the amount of defocusing increasing, the laser spot on the sample will be enlarged, the power density will be relatively more uniform, thus reducing the heat input at the weld center. The welding speed will affect the heat input in unit time, and the faster the speed, the smaller the heat input. Therefore, as shown in Fig. 7, with the same degree of variation to process parameters, the degree of influence on heat input: laser power > welding speed > amount of defocusing.

3.4. Molecular dynamics simulation

Molecular dynamics simulation (MD) is a method to simulate the time-related properties of the system with molecules and atoms as the basic particles, which has unique advantages for the study of dynamic diffusion at the atomic level. Due to the instantaneous nature of welding tests, the simulation at the atomic scale is implemented by molecular dynamics. In this paper, the process of weld formation is analyzed from atomic scale using molecular dynamics simulation. The **LAMMPS package** was used to model the three bonding stages (loading, unloading and cooling) of the Cu-Zn welding process with a model size of $5 \times 5 \times 70 \text{ nm}^3$. The parameters used in the MD simulations are shown in Table 4. The morphology of the interface, the atomic structure and the microscopic crystal structure were studied. **Ovito** software was used for visual analysis of the data generated in the simulation, and the addition process of each stage was analyzed, and the morphology of the combined interface of each stage was observed. The microstructure of each stage was studied by using common neighbor analysis (CNA) and centrosymmetry parameter (CSP). The diffusion behavior of atoms is closely related to pressure and temperature.

The Fig. 8-a shows the formation process of the welded joint. It can be seen that with the increase of laser energy, the welding process continues to occur. After the high-speed collision of Cu and Zn, the high-temperature excited atoms generated are active, but the diffusion is not obvious. With the welding process continuing, many disordered atomic structures are formed in the weld region, and when the welding process is over and stabilized, the disordered tissue structure is randomly distributed in the region of the weld center.

Corresponding analysis of the lattice structure in this part of the region, the lattice structure is transformed from the initial FCC phase to the BCC intermediate phase, and finally to the HCP phase. The atoms on the left and right sides gradually move closer together, and the content

of the atomic structure changes. At this stage, a large amount of BCC structure was produced, and the content of BCC structure changed twice with the change of temperature. This shows that a metastable BCC structure is generated during the condensation process of Zn atomic system at the faster cooling rate of welding. With the cooling process, the metastable BCC structure in the system is gradually transformed into a stable HCP structure. The results of simulation analysis confirm that many FCC's structures were transformed into HCP structures, which is the dislocation formation and evolution process of the welding process [35,36], as shown in Fig. 8-b. The nature of welding process dislocations is due to the formation of many irregular atomic arrangements, and these structures are randomly dispersed in the weld region, which strengthens the welded joint. The forming process of Cu-Zn weld was further verified by molecular dynamics simulation.

3.5. Mechanical property

3.5.1. Microhardness

The microhardness under different parameter combinations is shown in Fig. 9. The hardness of the whole welded joint is highest near the weld center. Especially, when the amount of defocusing in Fig. 9-c is 1 mm (No.6 parameters combination), the hardness reaches the maximum: 134 HV. A little further away from the centre of the weld, there would be a sudden drop in hardness which coincides with the position of the FZ, because the FZ consists mainly of coarse columnar crystals. Then the hardness would slowly recover to the base material level (125 HV), and this area of hardness recovery is the heat affected zone [42]. Combined with Fig. 7, it can be concluded that the distribution of the temperature field generated at different thermal inputs affects the magnitude and distribution of the welded joint hardness. Relatively high or low thermal input is not conducive to increasing hardness. Higher heat input reduces the zinc content, and lower heat input results in insufficient grain refinement. In addition, the higher heat.

3.5.2. Bending strength

The bending load–displacement curves of welded joints for different combinations of parameters and flexural strength are shown in Fig. 10. Combined with Fig. 7, it can be found that the bending properties of welded joints are close when the difference in heat input is small, such as the amount of defocusing. In contrast, when the heat input is widely varying, the bending properties of welded joints can vary considerably. In particular, the weld of 2000 W shown in Fig. 10-a has poor bending strength and ductility, and the weld of 1.5 mm shown in Fig. 10-b has the excellent bending properties.

During welding, the weld is inevitably depressed due to the evaporation of Zn, and it becomes a stress concentration region during bending, so the bending performance of the welded joint is worse than that of the base material [43]. When the heat input is in the appropriate range, the increase of heat input is conducive to the improvement of bending performance, because the increase of the heat input can refine the grain in the weld zone, and the reduction of Zn is also conducive to the improvement of ductility [44]. When the heat input exceeds a certain range, on the one hand, more Zn will be evaporated, resulting in deeper penetration, and its negative effect on the bending property will dominate. On the other hand, high temperature will also make the grains in the weld area become coarser, which will reduce the bending property of the welded joint.

In general, welded joints have weaker bending strength and ductility than the base material, but some samples have bending properties very close to the base material. Thus, welded joints with good bending properties can be obtained by controlling the heat input during welding.

4. Conclusions

Based on the above results and analysis, the following conclusions can be drawn:

Table 4
Parameters used in MD simulations [37–41].

Atomic	Parameters
Cu-Cu	EAM potential
Zn-Zn	MBMH potential: $A = 1.642 \times 10^{-9}$, $R = 0.29$, $Q = 4$, $\beta = 2.6$
Cu-Zn	LJ potential: $a = 0.025$, $R = 3.343$

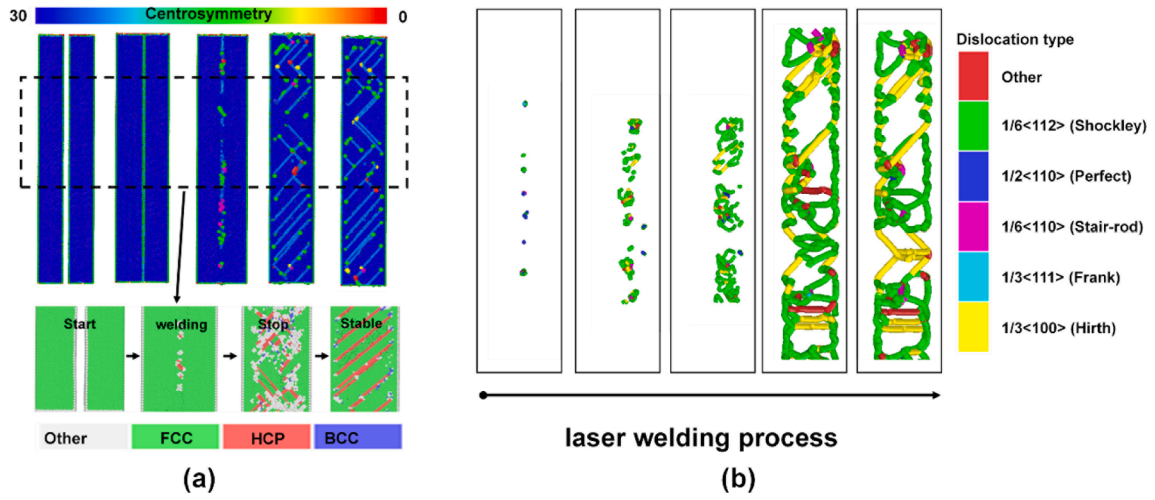


Fig. 8. Atomic scale simulation of welded joint formation process: (a) the transformation of lattice structure; (b) the formation and evolution of dislocation.

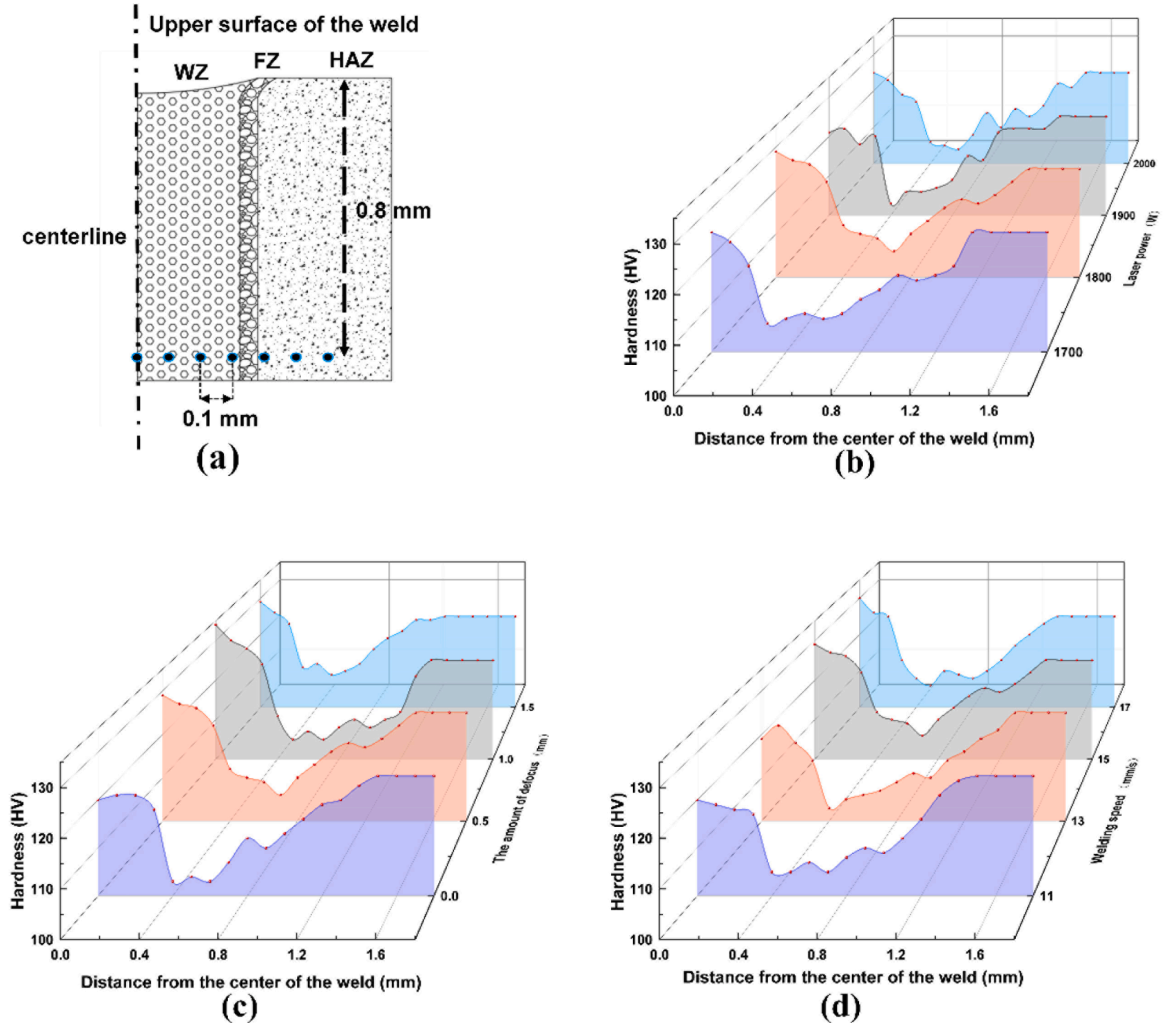


Fig. 9. (a) The schematic diagram of the position of the microhardness point on the weld; The relationship between hardness and (b) laser power, (c) the amount of defocusing and (d) welding speed input is accompanied by higher temperatures and a wider temperature range, which affects the hardness zone splitting.

(1) The welding quality of H62 brass laser butt joints is good, and no obvious pores, inclusions or cracks are found in most of the welded joints.

(2) The center of the WZ is mainly composed of equiaxed crystals. The FZ is dominated by columnar crystals. This is because different temperature gradients in each region result in different grain growth styles during the welding process. Then, the HAZ

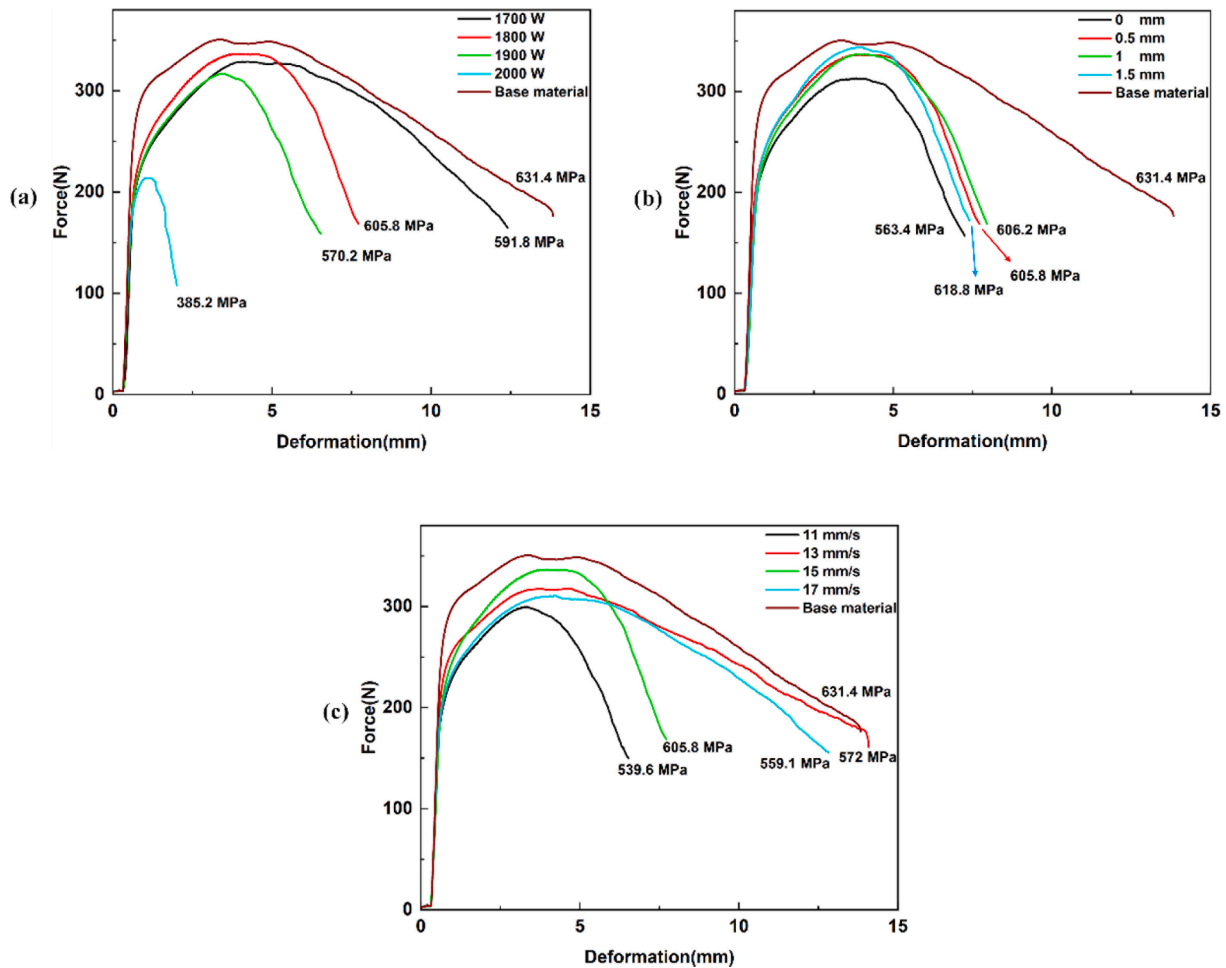


Fig. 10. Bending load-displacement curves of welded joints with different parameters: (a) laser power, (b) the amount of defocusing and (c) welding speed.

consists of α phase and β phase. And the β -phase is finely dispersed and forms fine crystal structure.

- (3) The macrosegregations were observed near both the top and back of the weld. The former is caused by insufficient heat input, resulting in low driving forces and the untimely escape of copper droplets and zinc vapour from the pores; the latter is caused by gravity and viscous forces after supercooling of the weld pool to the metastable miscibility gap in the Cu-Zn phase diagram.
- (4) In this study, the effect of different process parameters on the welding temperature field was studied by finite element simulations, and the simulation model can reflect the experiment well. In addition, molecular dynamics was used to describe the formation of welds at the atomic scale, providing microstructural insights for this study.
- (5) Temperature field distribution directly affects the magnitude and distribution of microhardness in welded joint. In addition, welded joints with good bending properties can also be obtained by controlling the heat input during welding. When the process parameters are No. 6 and No. 7, the welded joints have excellent microhardness and bending properties.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This study was supported by the Hubei Provincial Science and Technology Plan project, project number 2021BEC019. The authors express gratitude to Wuhan University of Technology's Material Research and Testing Center for its performance testing service.

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